FH Innovations Inc. (FHI)

Preliminary Phase

An Enterprise Architecture Proposal

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1. Executive Summary

FH Innovations Inc. (FHI) is an automotive services MSME with approximately 27 employees, specializing in decarbonizing, aircon treatment, engine treatment, and nitrogen tire inflation services delivered on-site at partner dealership service bays such as Toyota Balintawak and Toyota Manila Bay. FHI currently operates entirely on manual record-keeping — index cards and fragmented spreadsheets — with no dedicated IT infrastructure or IT personnel.

The central problem the architecture addresses is operational friction and working-capital compression caused by four gaps:

1. No reconciliation between items purchased and items delivered to dealer consignments
2. Recurring stockouts driven by the geographic distance of FHI's suppliers
3. Unsystematic Accounts Receivable (AR) aging and long-outstanding balances across multiple disconnected bank accounts
4. Manual, unverified pre-numbered Job Orders that create human error and untraceable cancellations.

The proposed To-Be architecture replaces these manual touchpoints with a lightweight, cloud-based Inventory & Invoicing / ERP solution built around five centralized data entities and three enforceable architecture principles.

2. Preliminary Phase — Framework and Principles

Boundary and Scope Statement

Dimension	Statement
Organizational units	FHI administration, finance and field service operations; touchpoints with partner dealership service bays
Geography	FHI headquarters and partner dealership bays across Metro Manila
Functional scope	Procurement, Consignment Inventory Tracking, Job Order Execution, Billing, AR Management

Exclusions	Changes to partner dealer systems (e.g., Toyota's Dealer Management Systems). Scope covers only FHI's interfaces, delivery handoffs and consignment ledgers
ADM scope	Preliminary to Phase D. Phases E/F are summarized as a roadmap (Section 9). Phases G/H are out of scope

Framework Interface Assessment

FHI has no formal IT department or software development lifecycle today. The EA framework therefore interfaces directly with existing business-administration rhythms rather than software:

- ADM Cycle Synchronization: ADM iterations are aligned with FHI's monthly financial closing and billing reconciliation cycle managed by Ms. Lisa (Administrator / Business Partner), so architecture milestones track the business's natural rhythm.
- Operational Integration: Transition milestones are phased to avoid disrupting daily decarbonizing and vehicle-treatment services at partner dealership bays. Change is introduced incrementally — starting with administrative data logging before expanding to real-time mobile technician tracking.

Tailored Enterprise Metamodel

FHI's lightweight operational structure calls for a streamlined subset of the standard TOGAF Foundation Metamodel, stripping out enterprise-grade constructs that do not apply to a flat, 27-person MSME:

Status	Entities
Retained	Business Actor, Business Role, Business Service, Business Process, Business Event, Data Object, Application Component, Application Service, Physical Technology Component, System Software, Location (FHI office plus each dealer bay, as a master data attribute only), Contract (consignment agreement terms and payment terms per dealer, as an attribute of Dealer & Partner Master)
Deleted	Logical Technology Component, Technology Service, enterprise-grade SLA constructs, Organization Unit hierarchy beyond FHI's flat structure

Tailored Content Framework

The following artifacts from the TOGAF Extended Content Framework will be produced during the corresponding ADM phases. They are referenced by name here and produced in full in Sections 4–6 of this document.

Phase B — Business Architecture artifacts

- Business Capability Map — core services: Decarbonizing, Aircon/Engine Treatment, Nitrogen Inflation.
- Process Flow Diagram (As-Is vs. To-Be) — Job Order issuance, dealer routing, and consignment reconciliation.
- Actor-to-Role Matrix — responsibilities across FHI administrative staff, service technicians, and dealer service advisors.

Phase C — Information Systems Architecture artifacts

- Data Entity / Data Flow Diagram — Purchase Order > Delivery Receipt > Consignment Usage > Billing Statement > AR Ledger.
- Application Communication Diagram — interaction between inventory logs, invoicing software, and multi-bank tracking.
- Information Use-Case Matrix — mapping data objects (Job Orders, POs, AR aging brackets) to business roles.

Phase D — Technology Architecture artifacts

- Technology Infrastructure Catalog — baseline administrative computers, field tablets, cloud backup systems.
- Network & Data Topology Diagram — secure cloud synchronization paths from field devices to administrative dashboards.

Principles Catalog

Principle 1 (P1) — Digital-First Over Manual Tracking

Statement: All business transactions, inventory movements, job orders, and billing metrics must be recorded digitally at the point of origin, replacing manual index cards and fragmented spreadsheets.

Rationale: Manual tracking introduces high rates of human error, misplaced records, and delayed billing reconciliations that directly compress FHI's working capital and create opaque operational visibility.

Implications: FHI staff must be equipped with mobile or desktop input tools; basic digital data-entry standards and training must be institutionalized across all 27 employees.

Principle 2 (P2) — Three-Way Consignment Reconciliation

Statement: Every item delivered to a partner dealership must maintain verifiable traceability across Purchase Orders, Delivery Receipts, and actual Jobs Done before billing generation.

Rationale: Current discrepancies between items bought and items delivered lead to uncollected revenue and inventory shrinkage across distributed dealer locations (e.g., Toyota Balintawak, Manila Bay).

Implications: Strict pre-numbered Job Order and delivery-receipt verification protocols, managed jointly by field staff and administration, so no service is rendered without digital logging.

Principle 3 (P3) — Automated Accounts Receivable Aging

Statement: Financial workflows must systematically track invoice issuance, collection windows, and aging brackets to trigger structured follow-up mechanisms automatically.

Rationale: Long outstanding payments severely impact FHI's cash flow, exacerbated by maintaining accounts across multiple disconnected banks and relying on reactive follow-up letters.

Implications: Adoption of centralized billing dashboards with automated aging alerts, scheduled collection windows, and streamlined bank-deposit reconciliation logs.